Transmission based precautions

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Definition

- Transmission-based precautions are the second tier of basic infection control. They are used for patients who are known or suspected to be infected or colonized with certain infectious agents that need **additional** control measures to prevent transmission.
- They are always used **in addition to** Standard Precautions, never instead of them.

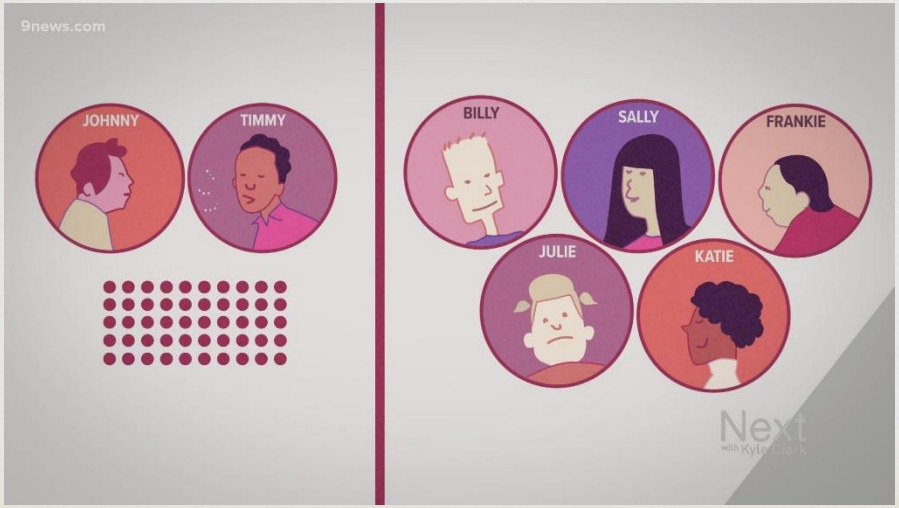
- Three types of transmission-based precautions have been developed: **contact, droplet and airborne.**
- Some infections need a combination of precautions (e.g., chickenpox and disseminated zoster: airborne + contact).

Contact precautions

- Used for known or suspected infections that spread by direct or indirect contact with the patient or the patient's environment (room surfaces, equipment and objects).
- Multidrug-resistant organisms (MDROs), colonized or infected: VRE, MRSA, ESBL, CRE, Candida auris.
- Other examples: C. difficile and norovirus (soap-and-water hand hygiene preferred), scabies, acute viral conjunctivitis.

Patient placement

- In a single-patient room if available.
- If not available: cohorting (patients with the same organism share a room).



PPE

- Gown and gloves
- for all interactions that may involve contact with the patient or the patient's environment
- Donning (put on): upon room entry
- Doffing (remove): before exiting the room, then hand hygiene



CONTACT PRECAUTIONS

GOWN + GLOVES



DON
put on

1. Clean hands
2. Gown
3. Gloves

DOFF
take off

1. Gloves
2. Gown
3. Clean hands
4. Exit

Put on before entering • Take off before leaving

* if needed (per risk of splash or combined precautions)

Based on CDC guidance. Follow your facility policy.

Patient-care equipment

- Use disposable or dedicated patient-care equipment (e.g., blood pressure cuffs).
- If common use of equipment for multiple patients is unavoidable, clean and disinfect it before use on another patient.

Transport and movement of patients

- Limit transport and movement to medically necessary purposes only.
- Cover or contain the infected or colonized areas of the patient's body.
- Remove and dispose of contaminated PPE and perform hand hygiene before leaving.
- Don clean PPE to handle the patient at the destination.



Cleaning and disinfection of the rooms

- Prioritize cleaning and disinfection: at least daily, or before use by another patient in outpatient settings.
- Focus on frequently touched surfaces and equipment near the patient; use a hospital-approved disinfectant (sporicidal for C. difficile).



Airborne precautions

- Prevent transmission of diseases spread by infectious respiratory particles that stay suspended in the air and are inhaled, even far from the source (traditionally droplet nuclei < 5 μm
- **Examples:** tuberculosis, measles, chickenpox and disseminated zoster, smallpox. COVID-19

Patient placement

- In an airborne infection isolation room (AIIR) with negative air pressure; mask the patient if they must leave the room.
- Limited resources: mask the patient in a private room with the door closed until transfer to an AIIR facility or return home.
- Restrict susceptible healthcare personnel from entering (measles, chickenpox, disseminated zoster, smallpox) if immune staff are available.

Airborne Infection Isolation Rooms

A

Positive Pressure Room

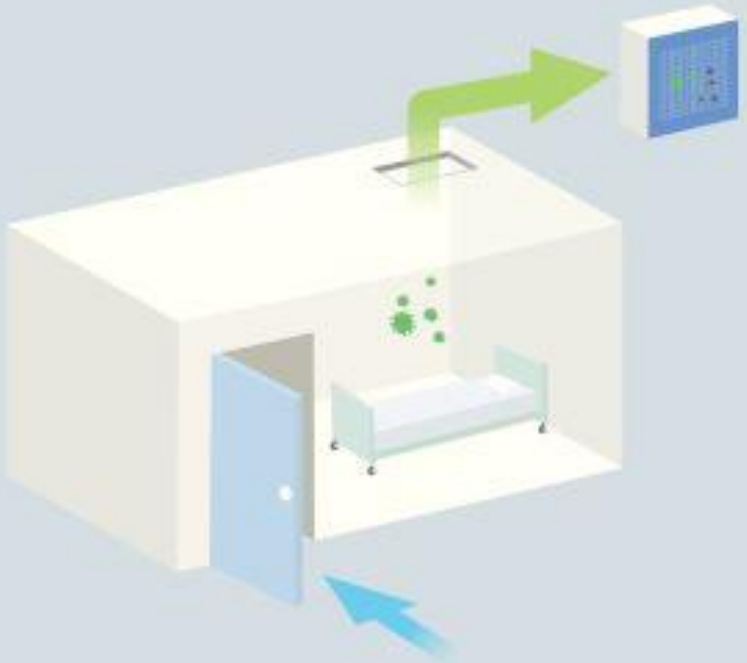


= Air enters the patient's room

= The air inside is push out, which is air contaminated with the virus from patient

B

Negative Pressure Room



= Air enters the patient's room

= Air forced suction out from the room and filter passed (antivirus)

PPE

- Wear a fit-tested NIOSH-approved N95 (or higher / equivalent FFP2) respirator before entering the patient's room.
- Remove the respirator after leaving the room and closing the door.
- Then perform hand hygiene.



HOW TO PROPERLY WEAR AN N95 MASK (RESPIRATOR)

CDC does not recommend that the general public wear N95 respirators to protect themselves from COVID-19. Only healthcare personnel with exposure to respiratory droplets should wear N95 respirators.

How to put on the N95 respirator (donning)



Wash your hands with soap and water or use an alcohol based hand rub



Hold the respirator in the palm of your hand with the straps facing the floor



Place the N95 respirator on your face covering your nose and mouth



Take the upper strap and put it behind your head towards the crown of your head



Pull the bottom strap up and over top of your head, and put it behind your head below your ears



Mold the nosepiece of the respirator over the bridge of your nose to obtain a tight seal



Perform a fit check (by blowing up air from your mouth and feeling it all around the mask with your fingers) to ensure there is a good seal against the skin

INFECTION PREVENTION AND CONTROL FOUNDATION

PROTECTOR Health & Hygiene

AIRBORNE
PRECAUTIONS

N95 RESPIRATOR



DON
put on

1. Clean hands
2. Gown*
3. N95
4. Eyewear*
5. Gloves

DOFF
take off

1. Gloves
2. Eyewear*
3. Gown*
4. Close door
5. N95
6. Clean hands

Fit-tested N95 • Remove it after the door is closed

* if needed (per risk of splash or combined precautions)

Based on CDC guidance. Follow your facility policy.

Transport and movement of patients

- Limit transport and movement to medically necessary purposes.
- Instruct the patient to wear a surgical mask and observe cough etiquette.
- Plan transport to limit exposure of others (e.g., no waiting in reception areas, empty elevator).
- Transport staff need no mask or respirator if the patient is masked and skin lesions are covered.



Cleaning and disinfection of the rooms

- Routine cleaning of high-touch surfaces is standard.
- Environmental services personnel should wear an N95 respirator on room entry.
- After the patient has left, the room should remain unoccupied for enough time to allow for complete air exchange (1 hour or less, depending on the air handling capacity of the facility).

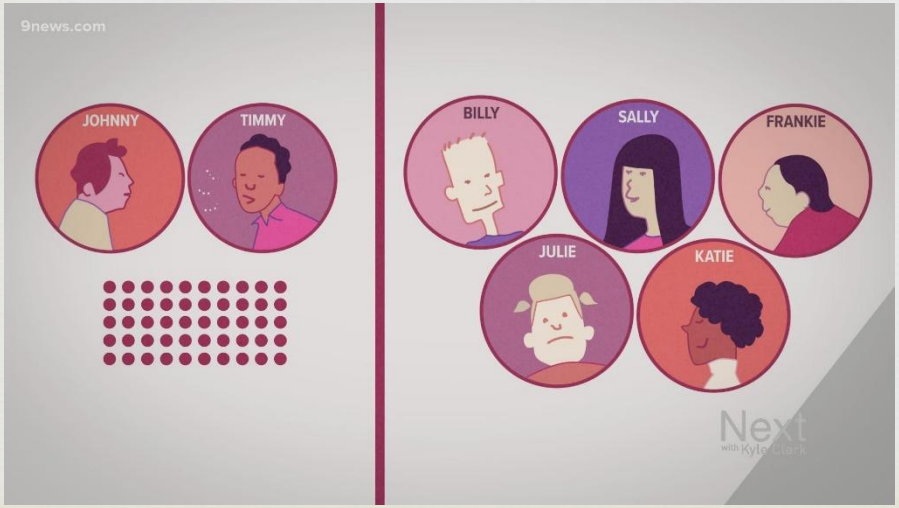


Droplet precautions

- Prevent transmission of diseases spread by large respiratory droplets (particles > 5 µm) that travel only short distances (about 1–2 m / 3–6 ft) before falling.
- They are produced when the infected patient talks, coughs, or sneezes, and during some procedures (e.g., suctioning, bronchoscopy).
- **Examples:** seasonal influenza, pertussis, mumps, rubella, adenovirus, rhinovirus, Neisseria meningitidis.

Patient placement

- A single-patient room is preferred; if unavailable, patients with the same infection may share a room.
- Separate beds by at least 1 m (3 ft); many facilities use 2 m (6 ft).



PPE

- Wear a surgical mask when entering the room; also put a mask on the patient.
- Add eye protection, gloves and gown per Standard Precautions if splash risk.
- For aerosol-generating procedures, wear an N95.



DROPLET
PRECAUTIONS

SURGICAL MASK



DON
put on

1. Clean hands
2. Gown*
3. Mask
4. Eyewear*
5. Gloves

DOFF
take off

1. Gloves
2. Eyewear*
3. Gown*
4. Exit
5. Mask
6. Clean hands

Mask on before entering • Mask off at the door

* if needed (per risk of splash or combined precautions)

Based on CDC guidance. Follow your facility policy.

Transport and movement of patients

- Limit the movement and transport of the patient to medically necessary purposes. The patient should wear a surgical mask during transport and follow respiratory hygiene and cough etiquette.

Cleaning and disinfection of the rooms

- Daily cleaning of high-touch and horizontal surfaces with a hospital-approved disinfectant.
- Environmental services personnel should don a surgical mask before room entry.



Radiology

- Notify the radiology department before the patient arrives; keep the isolation sign in place.
- Prefer portable imaging in the patient's room. If the patient must come, go straight in (no waiting area), ideally at the end of the list.
- Contact: gown and gloves, disposable cover for the detector or cassette, then clean and disinfect the machine, table and probes.
- Droplet: surgical mask for patient and staff. Airborne: masked patient, N95 for staff, room kept empty until the air has cleared.
- Hand hygiene before and after every patient; remove PPE before leaving the room.

Optometry

- Contact: adenoviral conjunctivitis spreads easily, so use gloves and hand hygiene, and never share eye-drop bottles.
- Clean slit-lamp chin and forehead rests, handles and other touched surfaces between patients; use a breath shield.
- Use single-use or disinfected tonometer tips, trial lenses and diagnostic contact lenses, per the manufacturer's instructions.
- Droplet: patient with cough or fever wears a surgical mask, stays 1–2 m from others, and non-urgent exams are postponed.
- See suspected infectious cases at the end of the session and clean the room afterwards.

Sources

- CDC. Transmission-Based Precautions, Infection Control guidance for healthcare providers (page reviewed April 2024).
- CDC/HICPAC. Guideline for Isolation Precautions: Preventing Transmission of Infectious Agents in Healthcare Settings (2007).
- WHO. Global technical consultation report on proposed terminology for pathogens that transmit through the air (April 2024).



Any questions?